

IMCA 221
Programming for Artists
Fall 2025

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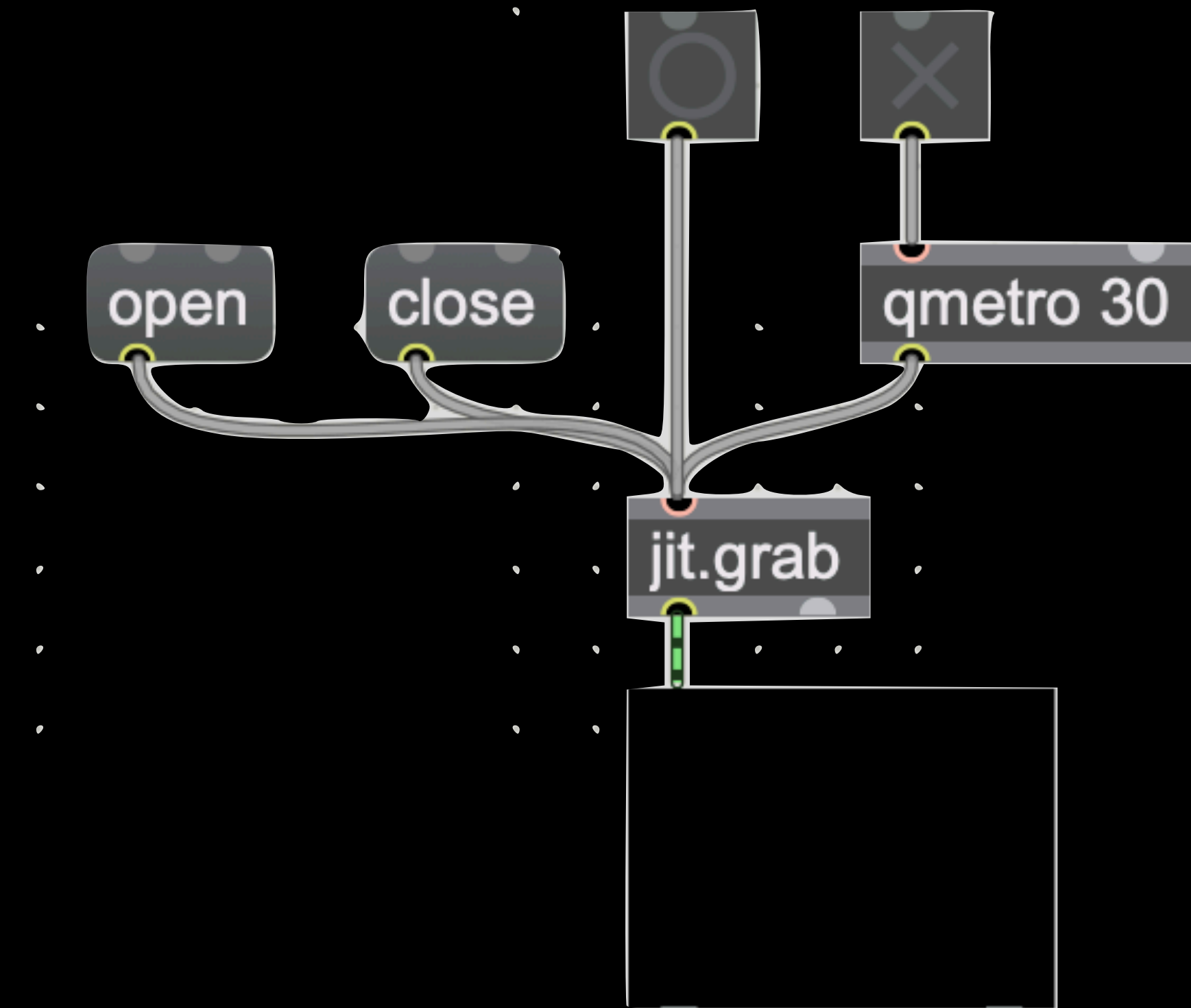
**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!

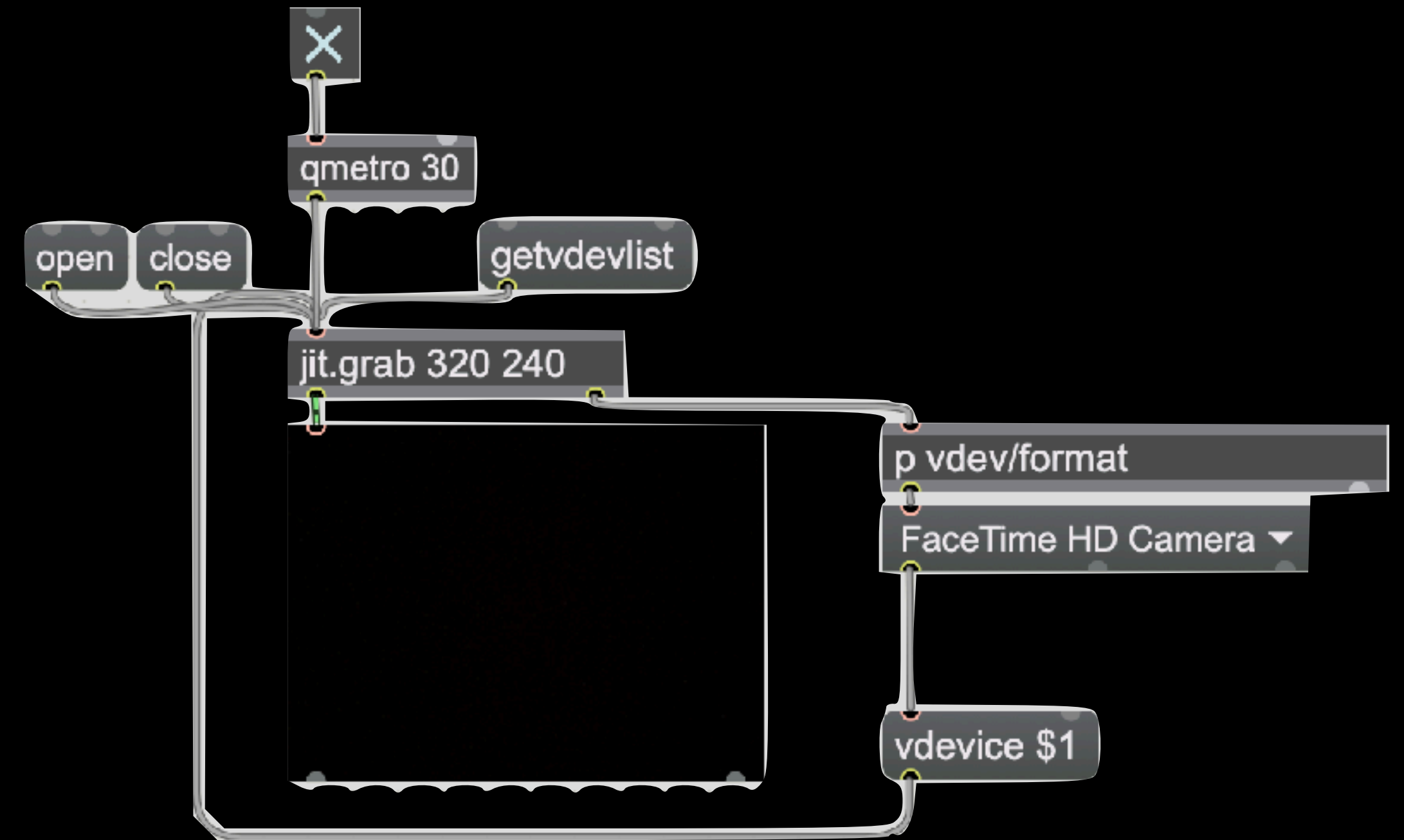
Using the camera
`jit.grab` is used to get
data from the camera.

This object needs an
open message and a
metro to take
continuous images

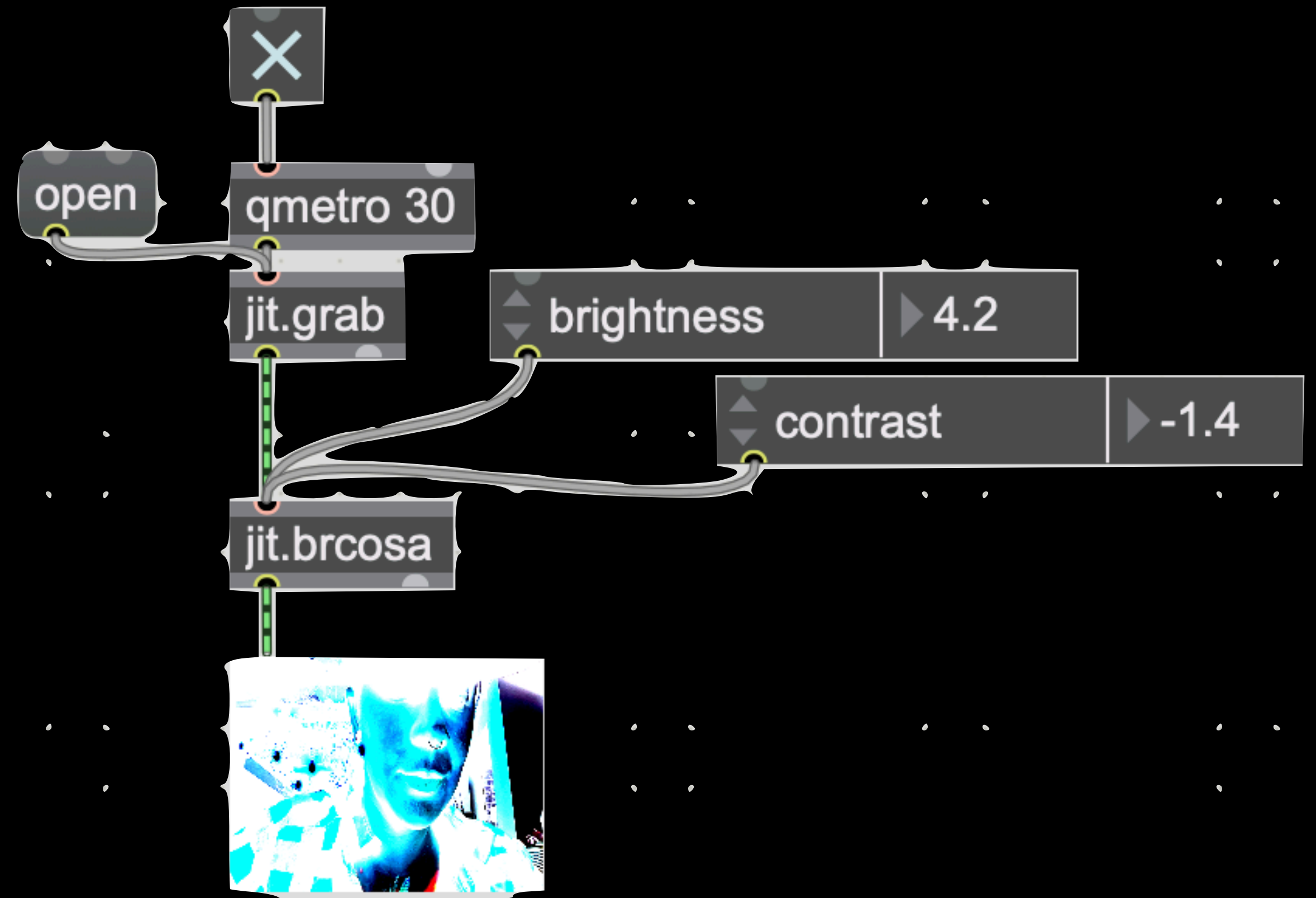


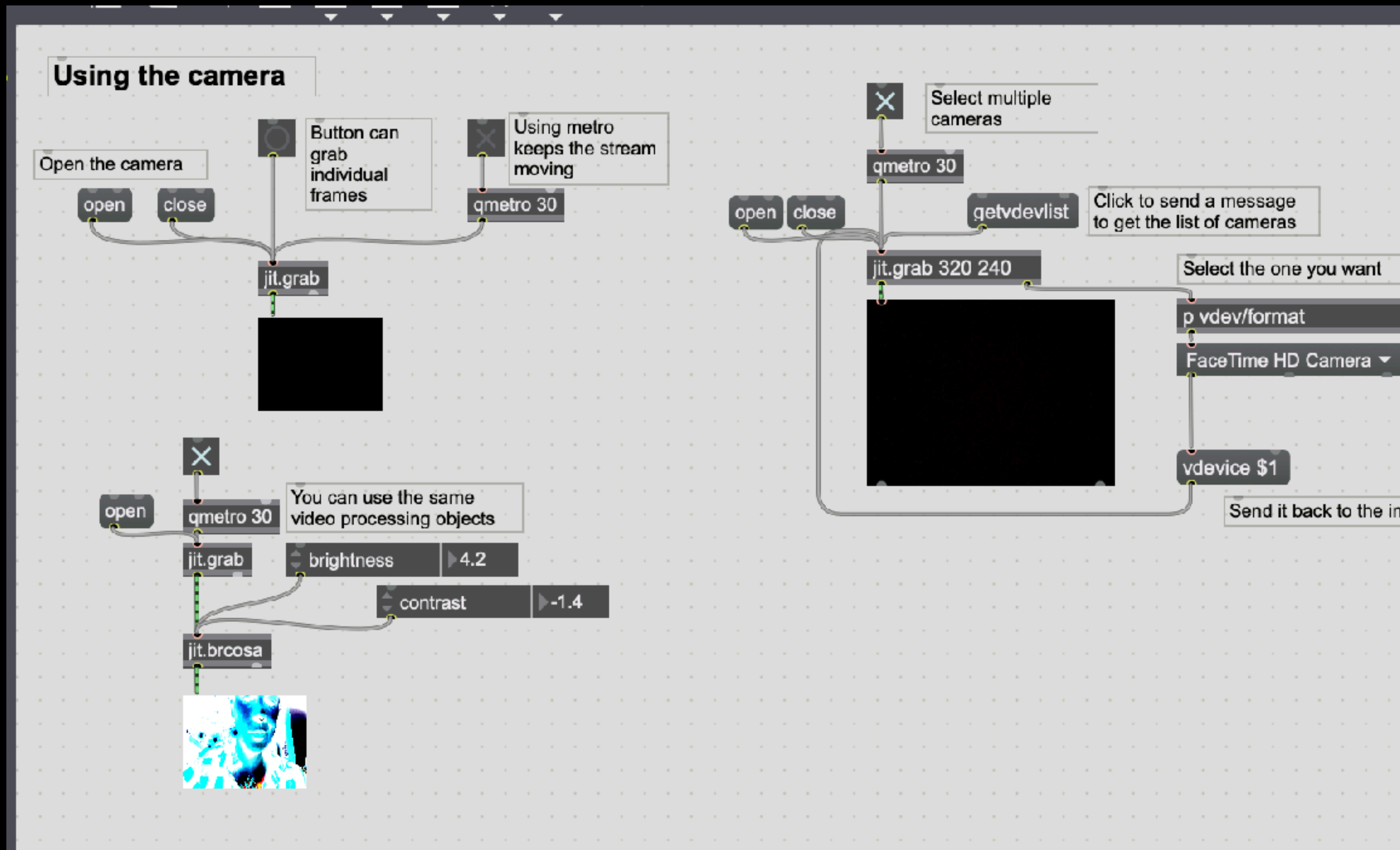
Using the camera

If you have multiple camera inputs you can select them using `getvdevlist`, find this example under the help menu for `jit.grab`



Using the camera
Use the same
video processing
objects





CameraCpature.maxpat

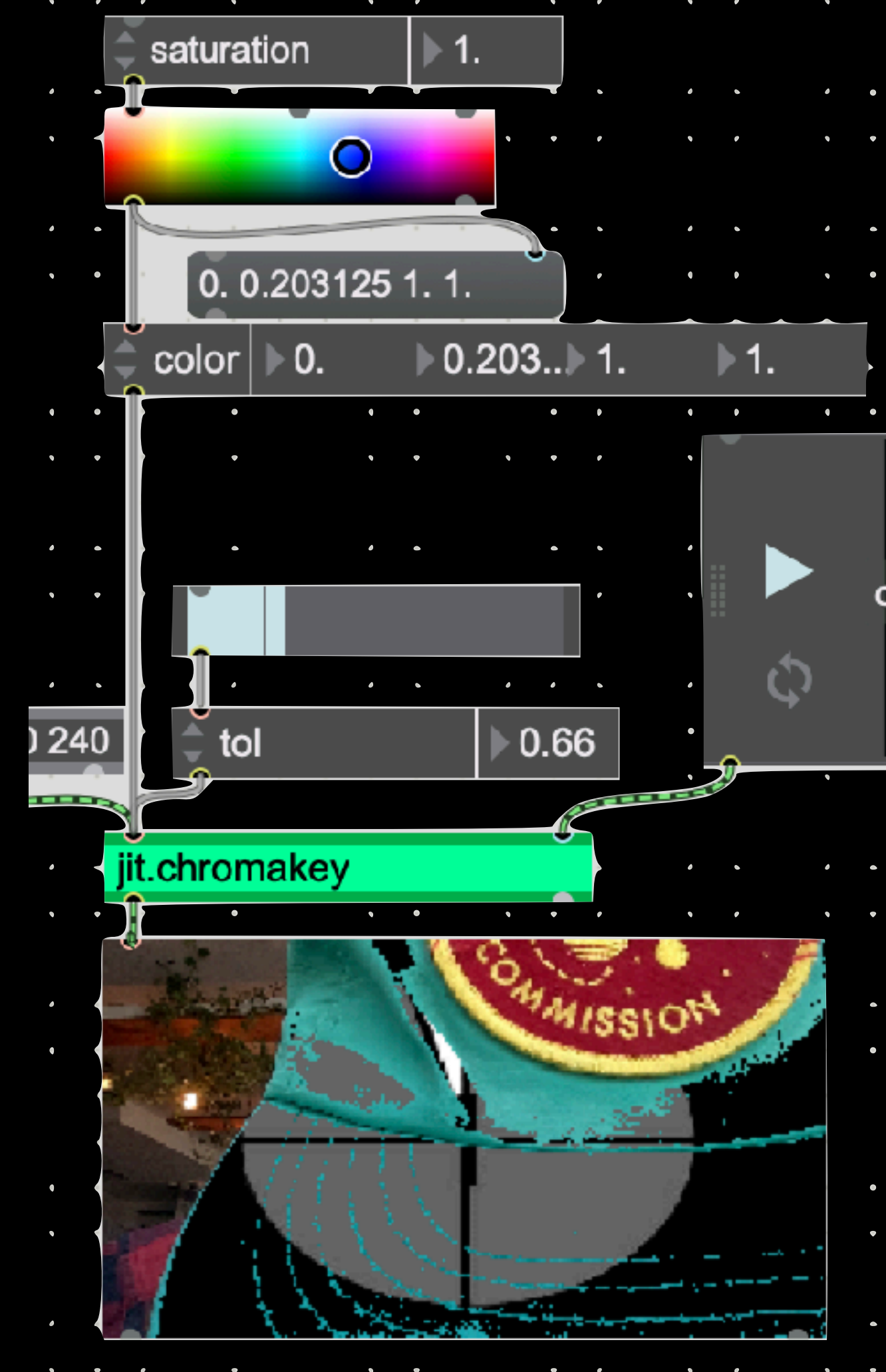
Chromakey

`jit.chromakey` is used to key out different colours. You can use a live feed or two videos or images. **Tolerance** attribute is a very basic way to control the key, but you'll want more control than that.



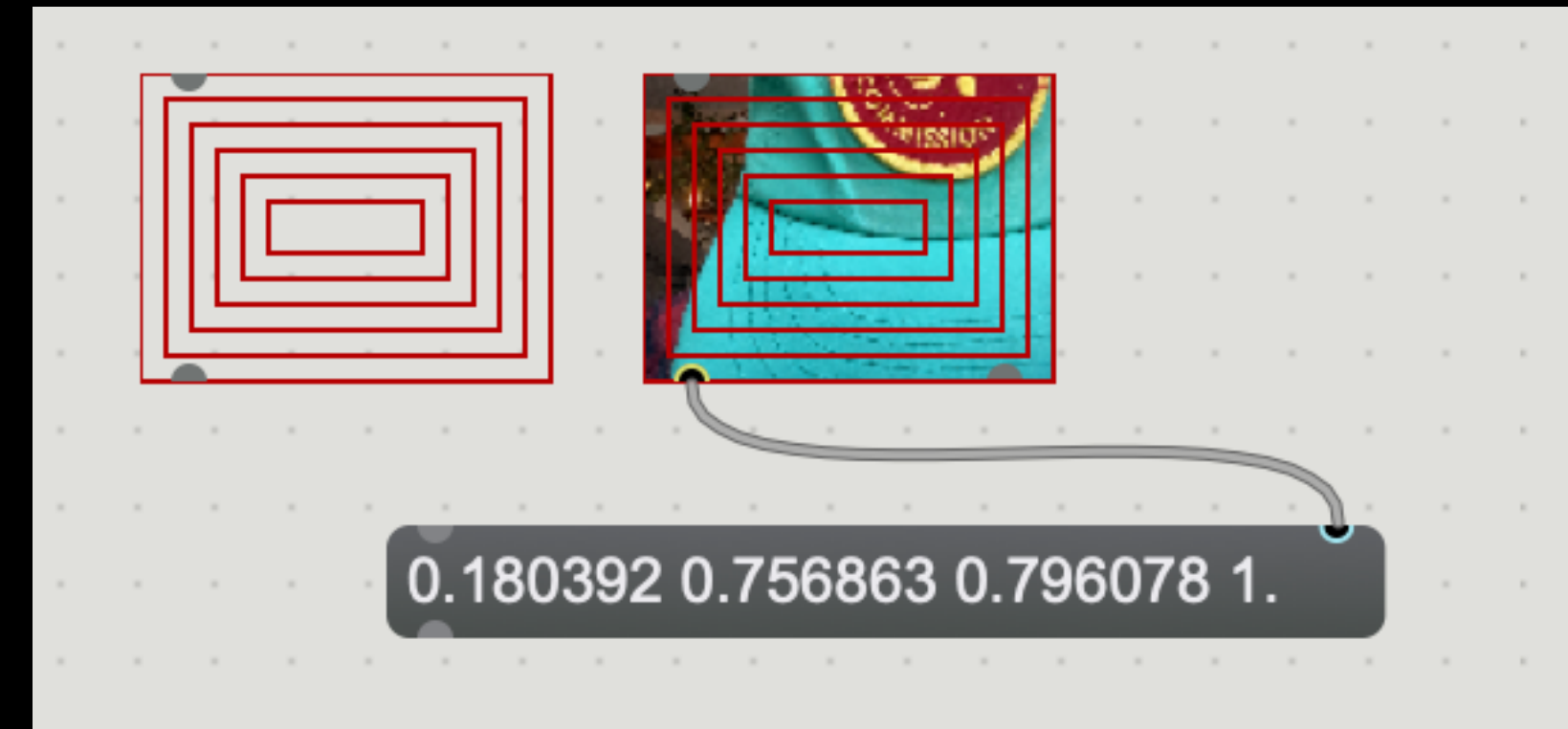
Chromakey

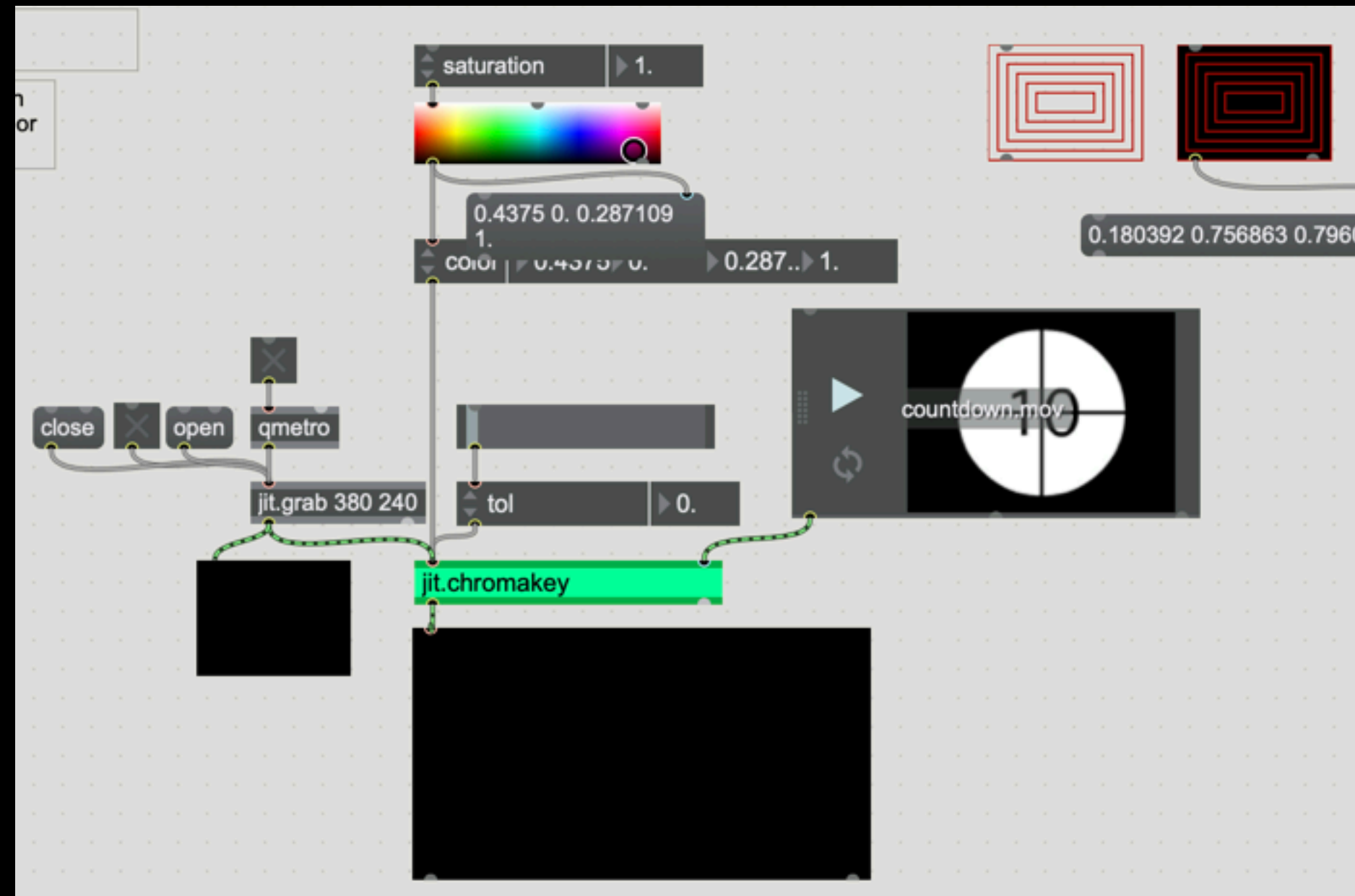
The **colour** attribute can be used to select which colour to key out. Manually input colours doesn't work very well, so we can use an object called **swatch**. Swatch outputs a list of (RGBA) of colour values.



Chromakey

We can also select a color directly from the video feed by using the suckah object. Suckah can be placed over anything (like a video feed) and will output the color value you clicked on as RGBA





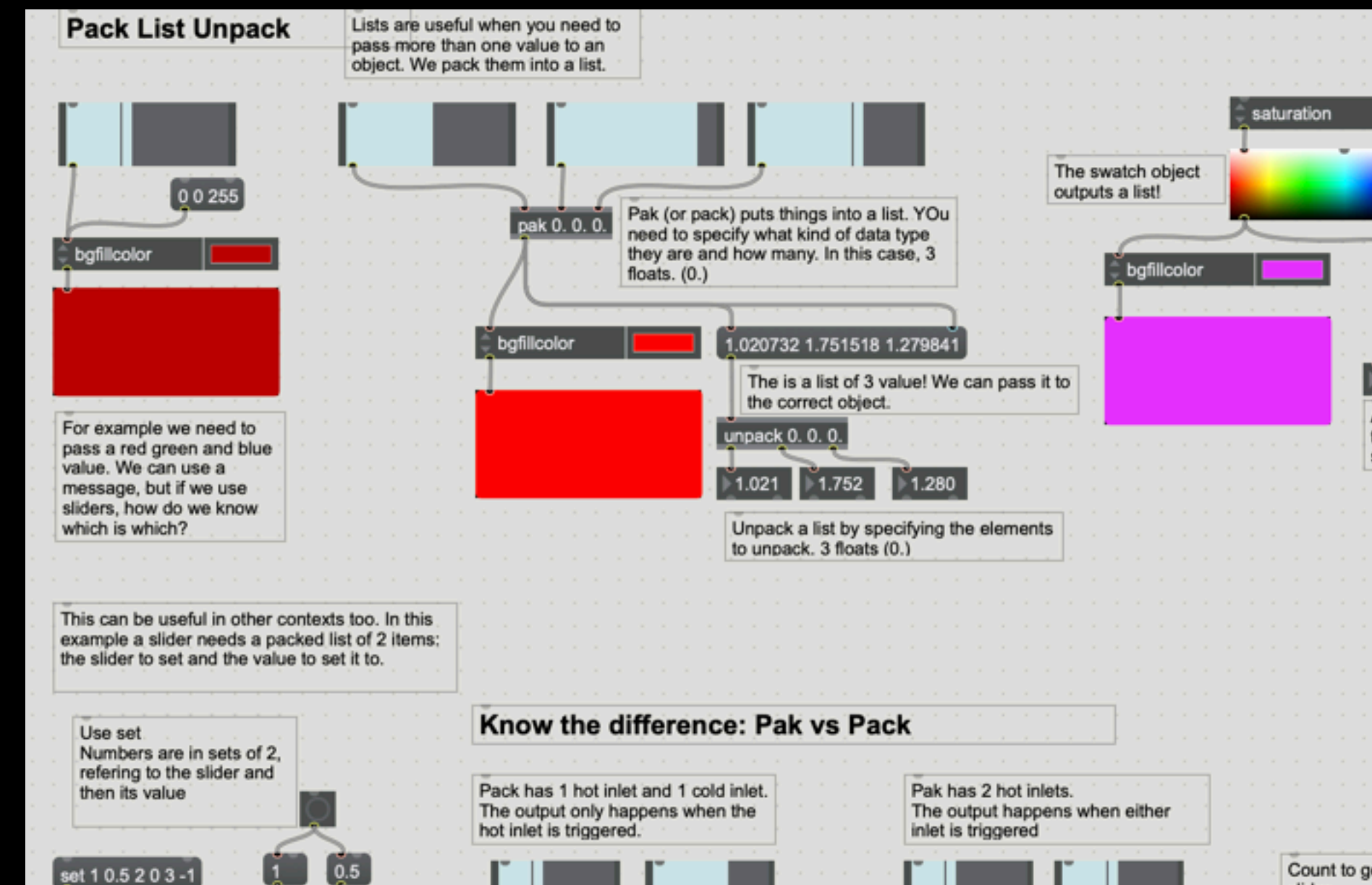
Chromakey.maxpat

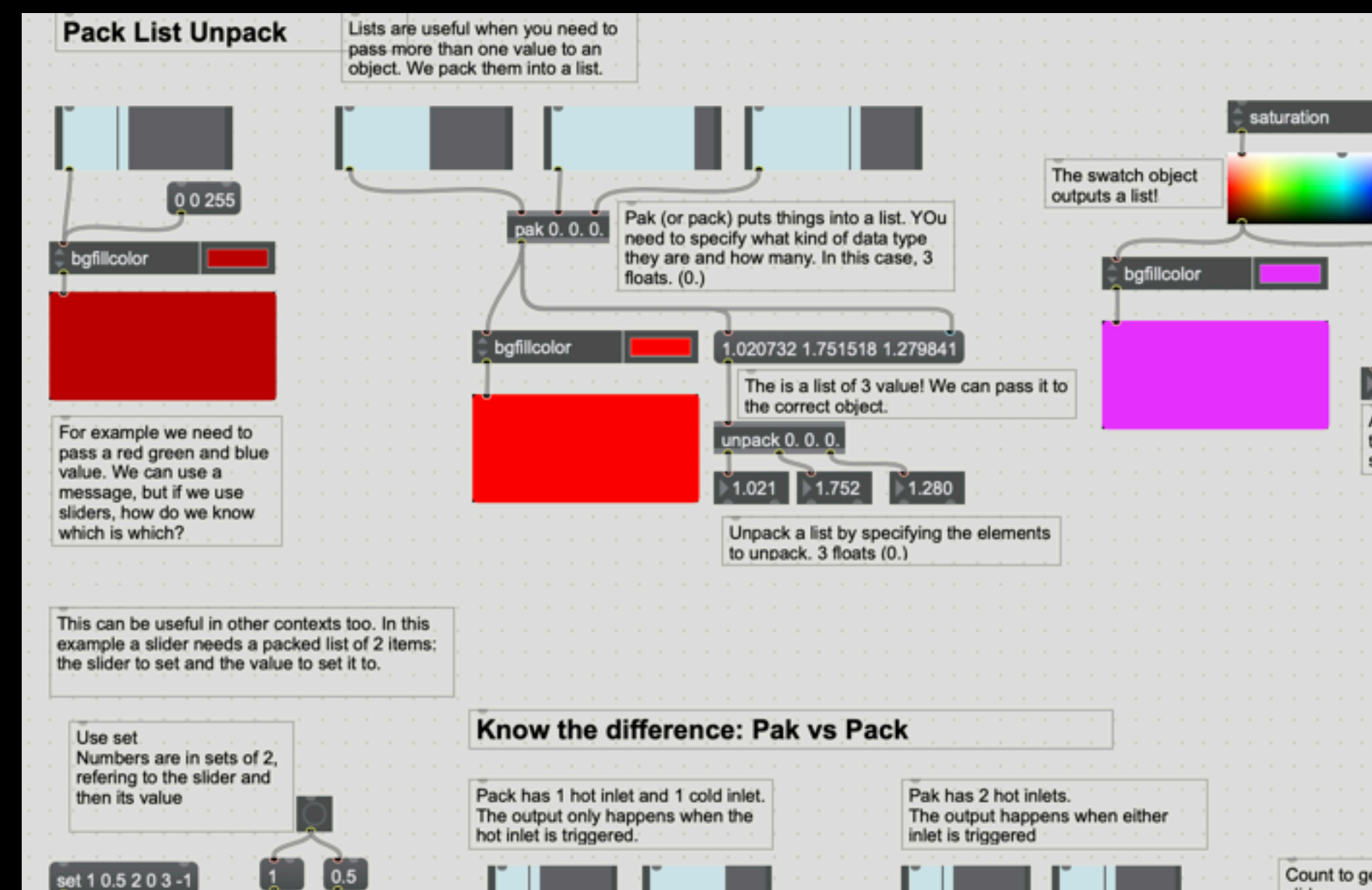
Lists, pack and unpack.

Lists: Combining data together into a list lets you pass a single message to an object

Pack/Pak: Assembles data into lists

Unpack: takes data out of lists





Lists, pack and unpack.

CV.Jit (Computer Vision)

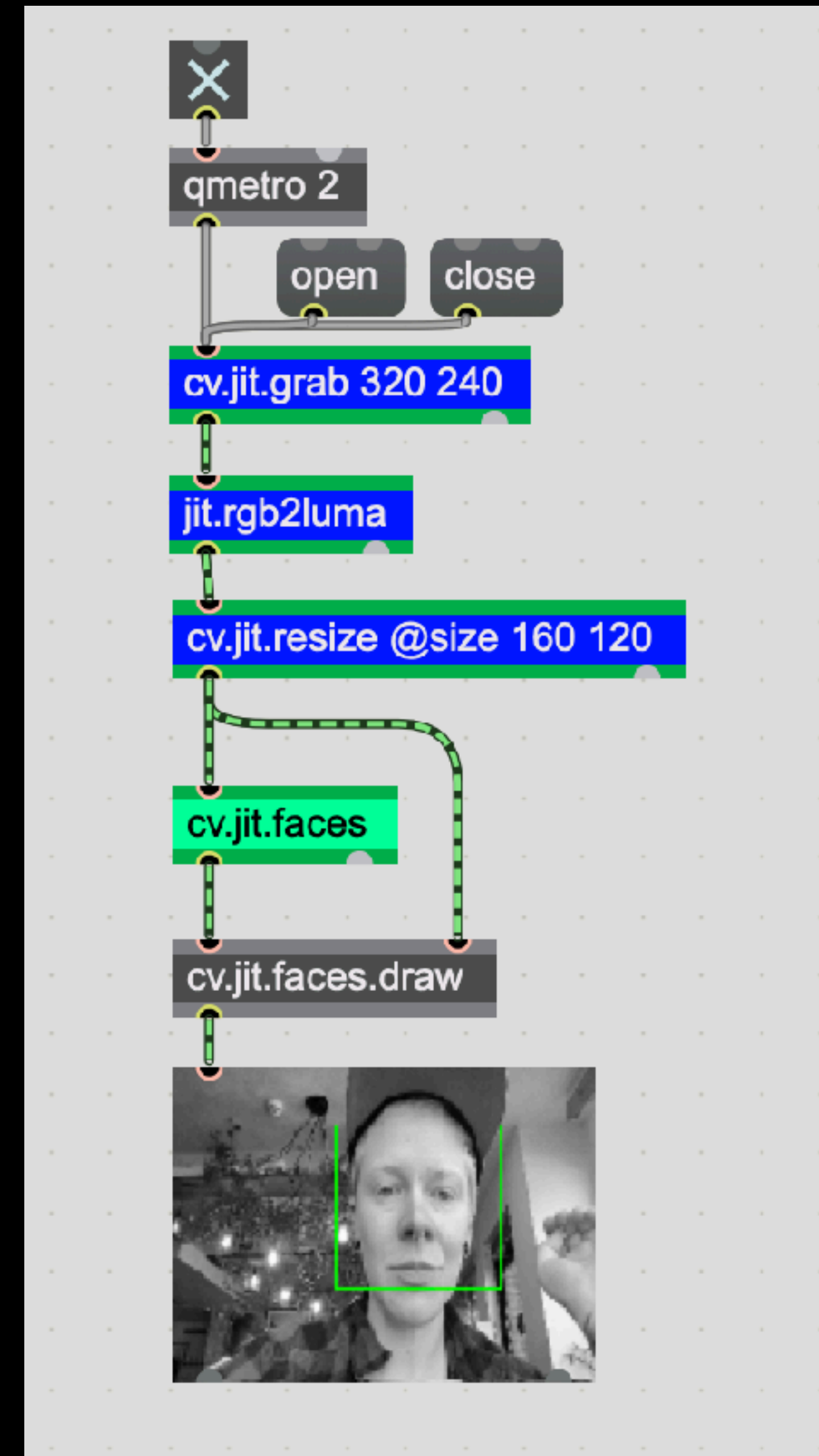
Download the cv.jit package. Open package manager, search for CV



CV.Jit (Computer Vision

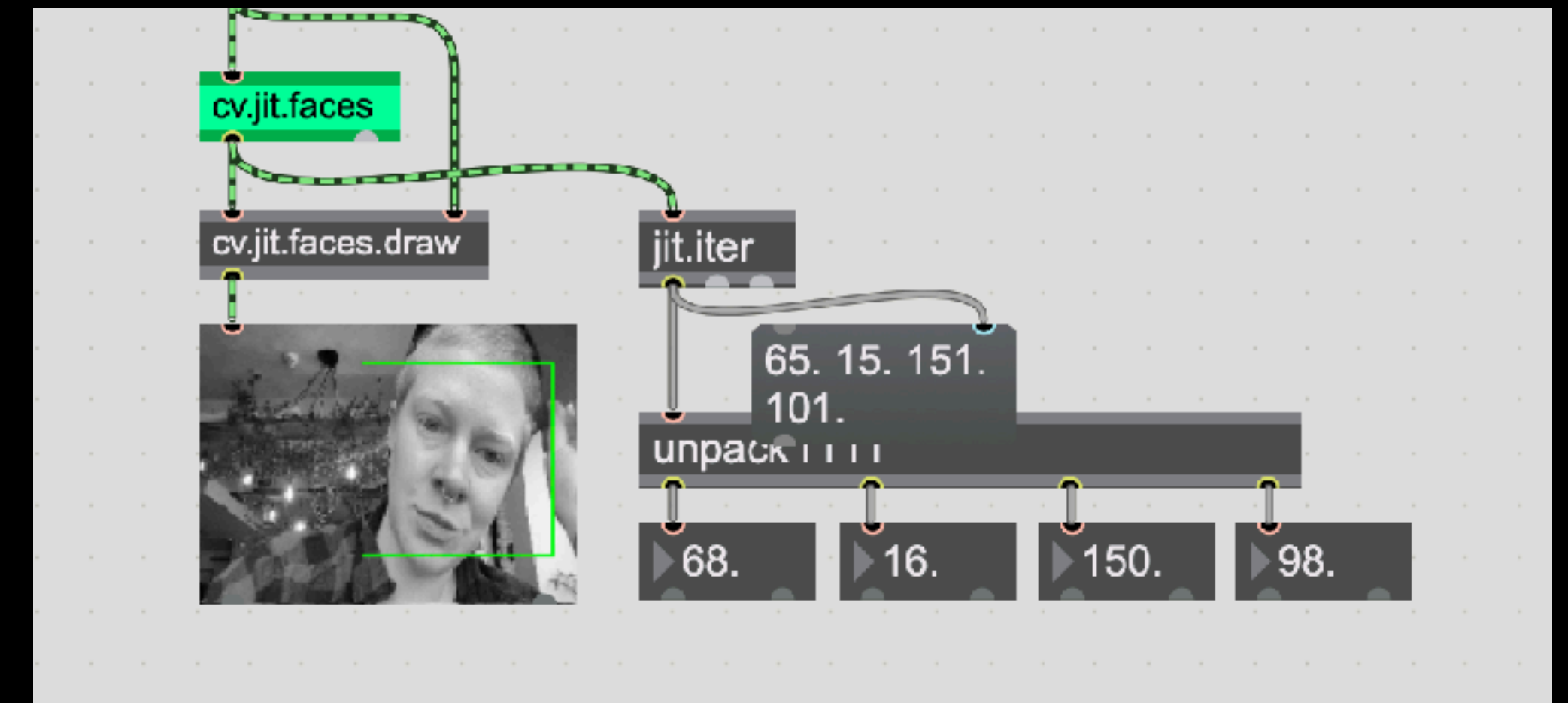
CV.jit does a lot of complex math and can be a bit slow. It requires a few objects to work properly. Cv objects start with cv.jit. Use the special cv.jit.grab object to capture video.

Translate to greyscale, reduce the size, use cv.jit.faces.



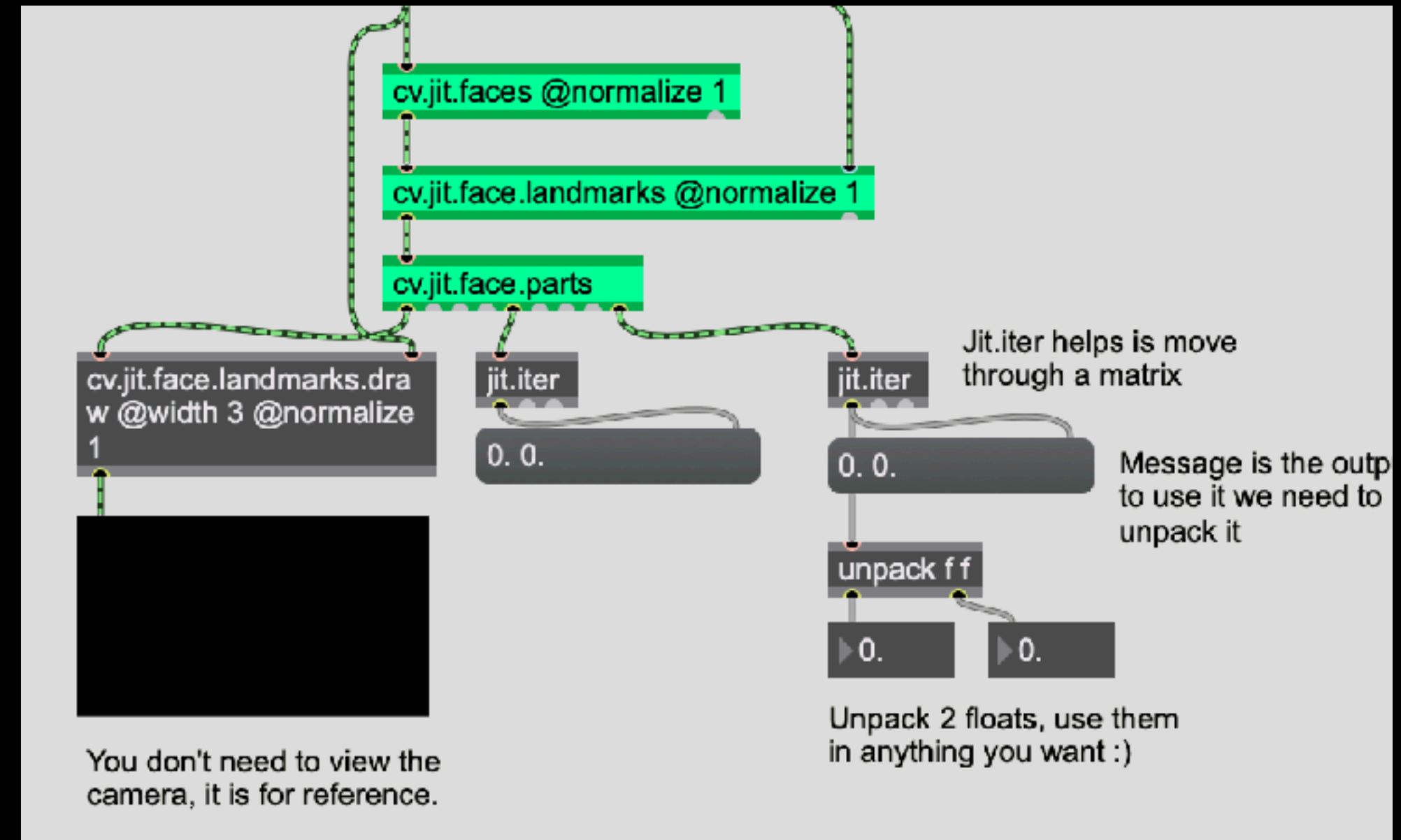
CV.Jit (Computer Vision)

Extract the values from `cv.jit.faces` using `jit.iter` and unpack the values for the bounding box.



CV.Jit (Computer Vision)

FacePartTracking.maxpat has an example of tracking the parts of the face. The `cv.jit.face.parts` object has outlets for x and y for various face parts.



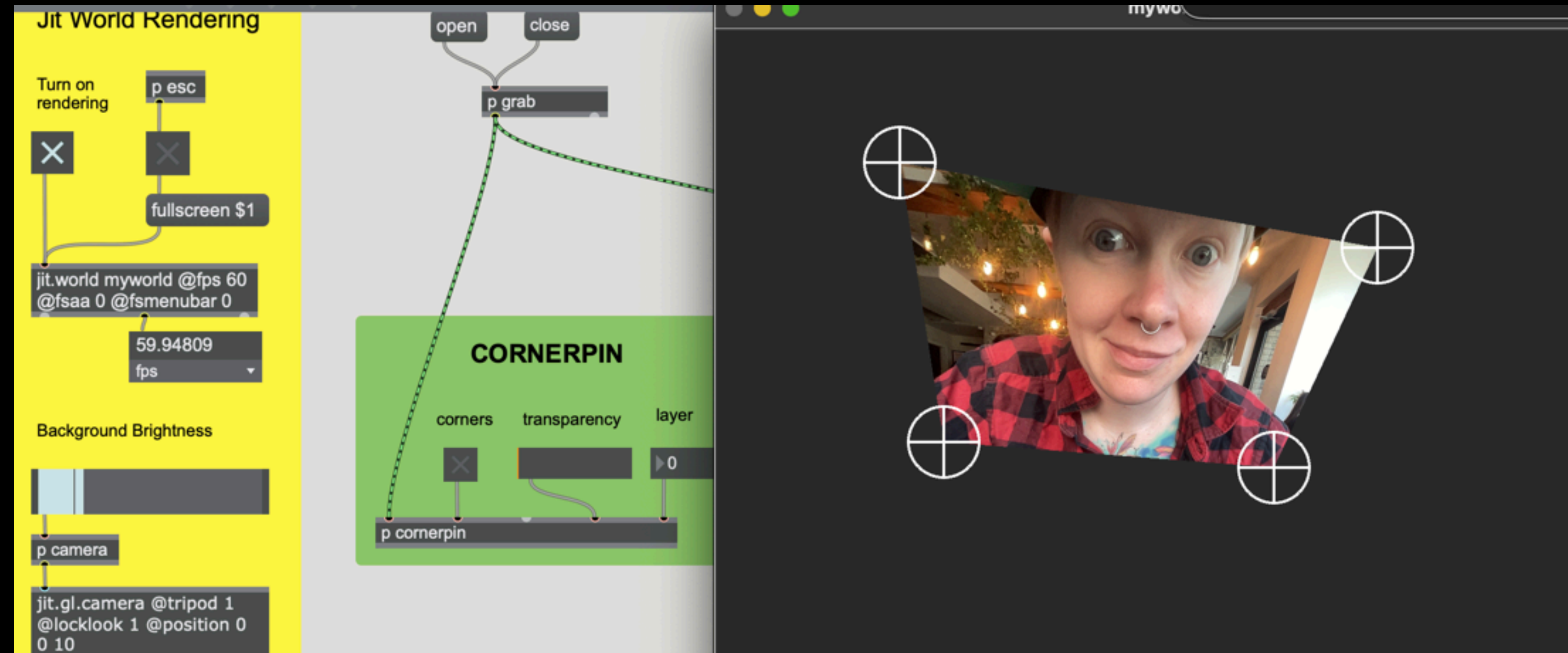
See a few other
examples in the
Computer Vision
Examples folder

OR under Extras >
CV.jit Object Guide

[cv.jit object guide](#)

blobs	cv.jit.blobs.bounds	Find blob bounds
tracking	cv.jit.blobs.centroids	Locate labeled blobs
faces	cv.jit.blobs.direction	Calculate blob direction
features	cv.jit.blobs.elongation	Calculate blob elongation
shape	cv.jit.blobs.moments	Analyse blob shapes
motion	cv.jit.blobs.orientation	Calculate blob axes
recognition	cv.jit.blobs.recon	Find shapes among blobs
lines	cv.jit.blobs.sort	Re-order labels for continuity
morphology	cv.jit.floodfill	Isolate a connected component in a binary image
filtering	cv.jit.label	Separate a binary image in blobs
perspective		
processing		
utility		
drawing		
new!		

Projection Mapping Example .maxpat



[https://
hcgilje.wordpress.
com/vpt/](https://hcgilje.wordpress.com/vpt/)



Video Experiment (midnight, after class)

- Explore audio tools we worked on in class
- Practice and explore tutorials online
- Create an experiment, which could be the beginning of a future project
- Your experiment should have a concept, even though it is not fully developed.

To hand in your project, use the project template on Moodle saved file (File > Save as Project) .maxpat

- A PDF that contains project documentation. See Project_Documentation_template
- A clear, strong image of your project.
- A screen capture of your max patch
- A link to video or audio recording of your project working (can be a video or a link to a private video on YouTube, Vimeo or Google)
- A 50-100 word explanation of your project inside your maxpatch Be sure to name files properly (no untitled-1.zip) All files are expected to be cleaned up and arranged in a reasonable, legible way. Videos should be clear, well light and show your project working.

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My Project Title

Lee Wilkins



Main Image:

Include a clear, strong image of your work. If the work is just audio, include an image of whatever context you imagine it in.

Artist Statement:

Write an artist statement that describes your work. Describe the work as though it is on a gallery wall, what would you read? 50-100 words. Remember, even a small project should have some kind of theme or concept. What do you want people to experience while viewing your work?

A good place to start is:

[Project Name] is an exploration of [concept]. By using [something, audio, video, samples from something?] viewers are able to experience [what do you want them to feel?].

Project Template on Moodle has everything you need, just replace the text and images with your own.

Video Experiment (midnight, after class)

25% Functionality / technical

25% creativity / concept

25% execution and quality

25% documentation

My Project Title

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**Download today's patch
on Moodle**

Midterm: Transform - Groups of 2-3, 25%, October 21

In this project you are invited to look beyond the computer screen and transform a space or atmosphere. Using primarily audio and video (and other if you want to) MaxMSP tools, bring us into a transformed within (or nearby) our classroom. This could mean changing the mood, forcing us to change perspectives, or imagine something new. Your project should **not** appear as a max patch, but as a piece for people to approach and engage with through a linear progression or interactive element. Your work should in some way have an arch or experience that builds or changes throughout the experience.

Be intentional. What do you want us to experience, and how are you bringing us there?

This work should feel complete - take note of where wires and cables are. Use projectors, lights, staging, curation etc to create a space. The work should be approaching gallery-ready state. Feel free to re-arrange any nearby space or reserve a critique room.

Homework:

Work on video Experiment (due next tonight)

Work on Midterm project
Next week: Midi and DMX